1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

High stromal-to-tumor ratio (TSR) creates a permissive stromal niche that augments AR–SOX2 transcriptional cooperation, promoting locus-specific remodeling of full-length transcript isoforms under endocrine therapy pressure. This isoform diversification, captured via Oases-enabled assembly of long-read–derived transcripts, drives lineage plasticity and functional escape from hormone blockade in AR-positive breast cancers. Clinically, tumors with high TSR and concurrent AR/SOX2 activity will exhibit reduced sensitivity to endocrine therapy (as measured by a therapy_resistance proxy or clinical surrogate) compared with TSR-low tumors lacking AR/SOX2 co-activation. Mechanistically, disrupting AR–SOX2 signaling or weakening stromal support will diminish full-length isoform diversity and restore endocrine sensitivity, indicating potential combinatorial therapeutic strategies.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	AR, SOX2
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

AR: AR–SOX2 cross-talk enables transcriptional reprogramming (isoform/locus-level changes) that under therapeutic pressure leads to lineage plasticity and persistent growth; why do some hormone-driven tumors fail endocrine therapy despite apparent AR expression? 2) Tier 1 path used: the AR ↔ SOX2 interaction and the link to locus/transcript-level rewiring (Oases/full-length transcripts) — these provide the mechanistic backbone that AR–SOX2 cooperation can generate alternative isoforms that drive resistance.

SOX2: SOX2 cross-talk enables transcriptional reprogramming (isoform/locus-level changes) that under therapeutic pressure leads to lineage plasticity and persistent growth; why do some hormone-driven tumors fail endocrine therapy despite apparent AR expression? 2) Tier 1 path used: the AR ↔ SOX2 interaction and the link to locus/transcript-level rewiring (Oases/full-length transcripts) — these provide the mechanistic backbone that AR–SOX2 cooperation can generate alternative isoforms that drive resistance.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Oases	Supporting modifier from knowledge graph

Secondary	full_length_transcripts	Supporting modifier from knowledge graph
Secondary	loci	Supporting modifier from knowledge graph
Secondary	enzalutamide	Supporting modifier from knowledge graph
Secondary	abiraterone	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: AR signaling and SOX2 function in transcriptional reprogramming can promote lineage plasticity and therapy resistance in hormone-driven cancers;

Prediction 2: the tumor stroma shapes cancer cell transcriptional programs via signaling, ECM cues, and paracrine factors;

Prediction 3: isoform- and locus-level transcript remodeling under therapeutic pressure can be revealed with long-read sequencing and assembly tools like Oases, enabling sensitive detection of full-length isoforms that may drive resistance.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Oases: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- full_length_transcripts: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.